



9/1/2026

N + - Network Fundamentals & Building Networks.

Wireless Networking.

TASK 1.

- ❖ List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

➤ Wi-Fi Standards:

<u>Wi-Fi Standard</u>	<u>Key Difference</u>
802.11n (Wi-Fi 4)	Works on 2.4 GHz and 5 GHz bands, with speeds up to 600 Mbps.
802.11ac (Wi-Fi 5)	Uses mainly the 5 GHz band and supports speeds up to 6.9 Gbps.
802.11ax (Wi-Fi 6)	Works on 2.4 GHz and 5 GHz bands and provides higher efficiency and speeds up to 9.6 Gbps.

TASK 2.

❖ Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

➤ Smartphone Connection to the Internet:

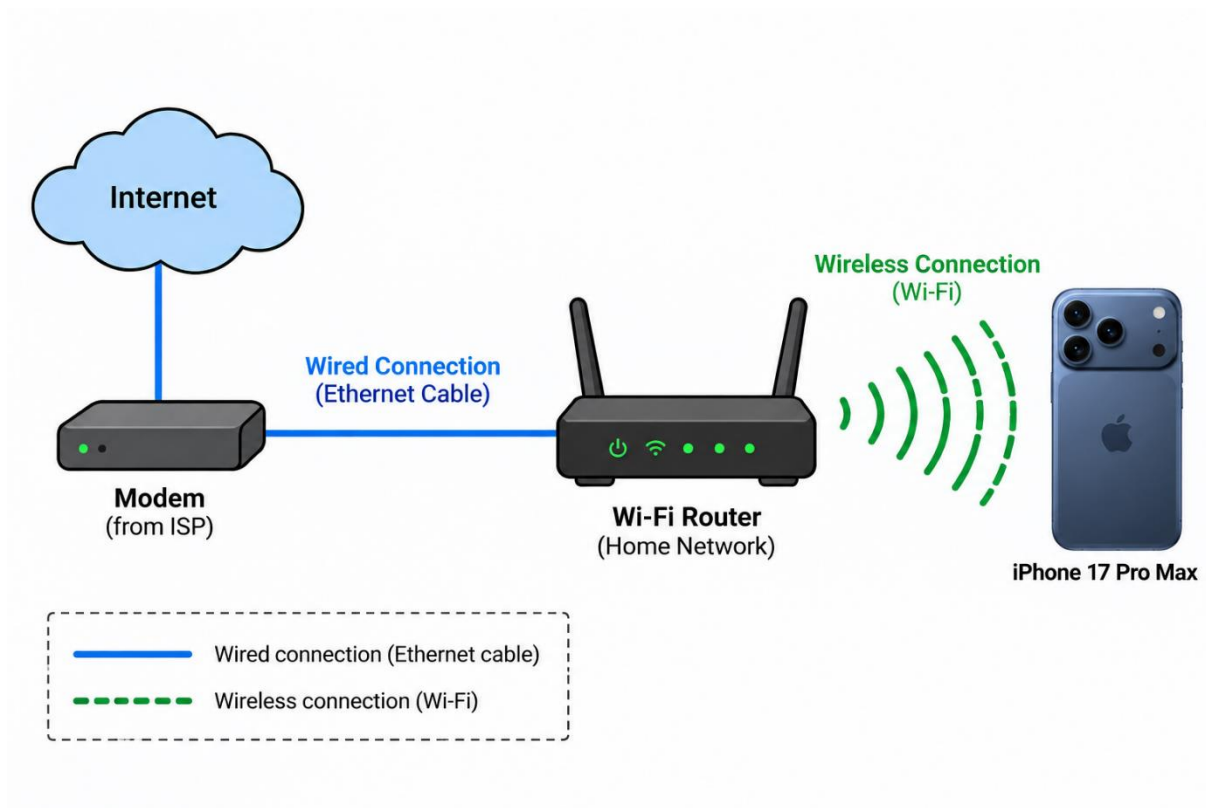


Figure : Smartphone Connecting to the Internet Through a Home Wi-Fi Router.

⌘ **Wired Part:** Internet → Modem → Wi-Fi Router (Ethernet Cable)

⌘ **Wireless Part:** Wi-Fi Router → iPhone 17 Pro Max (Wi-Fi)

TASK 3.

❖ Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why, Consider both signal strength and security when choosing.

➤ Wi-Fi Network Scan:

SSID	Signal Strength	Security Type
Home Network	Excellent (90%)	WPA3
College WiFi	Good (75%)	WPA2
Guest WiFi	Fair (55%)	WPA2

Network Chosen: Home Network

I would choose **Home Network** because it has the strongest signal and uses **WPA3 security**, providing a more secure and reliable connection.

TASK 4.

❖ Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

➤ SSID Hiding:

SSID hiding is a Wi-Fi feature that prevents the network name (SSID) from appearing in the list of available Wi-Fi networks. Users must enter the network name manually to connect.

Advantage: It makes the Wi-Fi network less visible to nearby users.

Disadvantage: It does not provide strong security, because the hidden SSID can still be discovered using network-scanning tools.